



Literature reviews as independent studies: guidelines for academic practice

Sascha Kraus^{1,2} · Matthias Breier³ · Weng Marc Lim^{4,8,22} · Marina Dabić^{5,6} · Satish Kumar^{7,8} · Dominik Kanbach^{9,10} · Debmalya Mukherjee¹¹ · Vincenzo Corvello¹² · Juan Piñeiro-Chousa¹³ · Eric Liguori¹⁴ · Daniel Palacios-Marqués¹⁵ · Francesco Schiavone^{16,17} · Alberto Ferraris^{18,21} · Cristina Fernandes^{19,20} · João J. Ferreira¹⁹

Received: 15 August 2022 / Accepted: 7 September 2022 / Published online: 14 October 2022
© The Author(s) 2022

Abstract

Review articles or literature reviews are a critical part of scientific research. While numerous guides on literature reviews exist, these are often limited to the philosophy of review procedures, protocols, and nomenclatures, triggering non-parsimonious reporting and confusion due to overlapping similarities. To address the aforementioned limitations, we adopt a pragmatic approach to demystify and shape the academic practice of conducting literature reviews. We concentrate on the types, focuses, considerations, methods, and contributions of literature reviews as independent, standalone studies. As such, our article serves as an overview that scholars can rely upon to navigate the fundamental elements of literature reviews as standalone and independent studies, without getting entangled in the complexities of review procedures, protocols, and nomenclatures.

Keywords Literature reviews · Bibliometrics · Meta Analysis · Methods · Contributions

JEL classification M1 · M10 · M19 · M20

1 Introduction

A literature review – or a review article – is “a study that analyzes and synthesizes an existing body of literature by identifying, challenging, and advancing the building blocks of a theory through an examination of a body (or several bodies) of prior work (Post et al. 2020, p. 352). Literature reviews as standalone pieces of work may allow

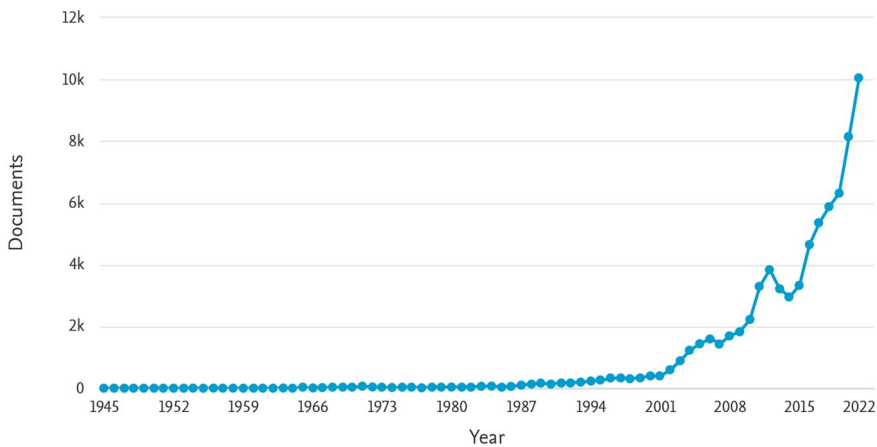


Fig. 1 Full-year publication trend of review articles on Scopus (1945–2021)

researchers to enhance their understanding of prior work in their field, enabling them to more easily identify gaps in the body of literature and potential avenues for future research. More importantly, review articles may challenge established assumptions and norms of a given field or topic, recognize critical problems and factual errors, and stimulate future scientific conversations around that topic. Literature reviews¹ come in many different formats and purposes:

- Some review articles conduct a critical evaluation of the literature, whereas others elect to adopt a more exploratory and descriptive approach.
- Some reviews examine data, methodologies, and findings, whereas others look at constructs, themes, and theories.
- Some reviews provide summaries by holistically synthesizing the existing research on a topic, whereas others adopt an integrative approach by assessing related and interdisciplinary work.

The number of review articles published as independent or standalone studies has been increasing over time. According to Scopus (i.e., *search database*), reviews (i.e., *document type*) were first published in journals (i.e., *source type*) as independent studies in 1945, and they subsequently appeared in three digits yearly from the late 1980s to the late 1990s, four digits yearly from the early 2000s to the late 2010s, and five digits in the year 2021 (Fig. 1). This increase is indicative that reviewers and editors in business and management research alike see value and purpose in review articles to such a level that they are now commonly accepted as independent, standalone studies. This development is also reflected in the fact that some academic journals exclusively publish review articles (e.g., the *Academy of Management Annals*, or the *International Journal of Management Reviews*), and journals publishing in

¹ Our focus here is on standalone literature reviews in contrast with literature reviews that form the theoretical foundation for a research article.

various fields often have special issues dedicated to literature reviews on certain topic areas (e.g., the *Journal of Management* and the *Journal of International Business Studies*).

One of the most important prerequisites of a high-quality review article is that the work follows an established methodology, systematically selects and analyzes articles, and periodically covers the field to identify latest developments (Snyder 2019). Additionally, it needs to be reproducible, well-evidenced, and transparent, resulting in a sample inclusive of all relevant and appropriate studies (Gusenbauer and Hadaway 2020; Hansen et al. 2021). This observation is in line with Palmatier et al. (2018), who state that review articles provide an important synthesis of findings and perspectives in a given body of knowledge. Snyder (2019) also reaffirmed this rationale, pointing out that review articles have the power to answer research questions beyond that which can be achieved in a single study. Ultimately, readers of review articles stand to gain a one-stop, state-of-the-art synthesis (Lim et al. 2022a; Popli et al. 2022) that encapsulates critical insights through the process of re-interpreting, re-organizing, and re-connecting a body knowledge (Fan et al. 2022).

There are many reasons to conduct review articles. Kraus et al. (2020) explicitly mention the benefits of conducting *systematic* reviews by declaring that they often represent the first step in the context of larger research projects, such as doctoral dissertations. When carrying out work of this kind, it is important that a holistic overview of the current state of literature is achieved and embedded into a proper synthesis. This allows researchers to pinpoint relevant research gaps and adequately fit future conceptual or empirical studies into the state of the academic discussion (Kraus et al., 2021). A review article as an independent or standalone study is a viable option for any academic – especially young scholars, such as doctoral candidates – who wishes to delve into a specific topic for which a (recent) review article is not available.

The process of conducting a review article can be challenging, especially for novice scholars (Boell and Cecez-Kecmanovic 2015). Therefore, it is not surprising that numerous guides have been written in an attempt to improve the quality of review studies and support emerging scholars in their endeavors to have their work published. These guides for conducting review articles span a variety of academic fields, such as engineering education (Borrego et al. 2014), health sciences (Cajal et al. 2020), psychology (Laher and Hassem 2020), supply chain management (Durach et al. 2017), or business and entrepreneurship (Kraus et al. 2020; Tranfield et al. 2003) – the latter were among the first scholars to recognize the need to educate business/management scholars on the roles of review studies in assembling, ascertaining, and assessing the intellectual territory of a specific knowledge domain. Furthermore, they shed light on the stages (i.e., planning the review, conducting the review, reporting, and dissemination) and phases (i.e., identifying the need for a review, preparation of a proposal for a review, development of a review protocol, identification of research, selection of studies, study quality assessment, data extraction and monitoring progress, data synthesis, the report and recommendations, and getting evidence into practice) of conducting a systematic review. Other scholars have either adapted and/or developed new procedures (Kraus et al. 2020; Snyder 2019) or established review protocols such as the *Preferred Reporting Items for Systematic Reviews and*

Meta-Analyses (PRISMA) flow diagram (Moher et al. 2015). The latter provides a checklist that improves transparency and reproducibility, thus reducing questionable research practices. The declarative and procedural knowledge of a checklist allows users to derive value from (and, in some cases, produce) methodological literature reviews.

Two distinct and critical gaps or issues provide impetus for our article. First, while the endeavors of the named scholars are undoubtedly valuable contributions, they often encourage other scholars to explain the methodology of their review studies in a non-parsimonious way (*1st issue*). This can become problematic if this information distracts and deprives scholars from providing richer review findings, particularly in instances in which publication outlets impose a strict page and/or word limit. More often than not, the early parts (i.e., stages/phases, such as needs, aims, and scope) of these procedures or protocols are explained in the introduction, but they tend to be reiterated in the methodology section due to the prescription of these procedures or protocols. Other parts of these procedures or protocols could also be reported more parsimoniously, for example, by filtering out documents, given that scientific databases (such as *Scopus* or *Web of Science*) have since been upgraded to allow scholars to select and implement filtering criteria when conducting a search (i.e., criterion-by-criterion filtering may no longer be necessary). More often than not, the procedures or protocols of review studies can be signposted (e.g., bracket labeling) and disclosed in a sharp and succinct manner while maintaining transparency and replicability.

Other guides have been written to introduce review nomenclatures (i.e., names/naming) and their equivalent philosophical underpinnings. Palmatier et al. (2018) introduced three clearly but broadly defined nomenclatures of literature reviews as independent studies: domain-based reviews, theory-based reviews, and method-based reviews. However, such review nomenclatures can be confusing due to their overlapping similarities (*2nd issue*). For example, Lim et al. (2022a) highlighted their observation that the review nomenclatures associated with domain-based reviews could also be used for theory-based and method-based reviews.

The two aforementioned issues – i.e., *the lack of a parsimonious understanding and the reporting of the review methodology*, and *the confusion emerging from review nomenclatures* – are inarguably the unintended outcomes of diving into an advanced (i.e., higher level) understanding of literature review procedures, protocols, and nomenclatures from a philosophical perspective (i.e., underpinnings) without a foundational (i.e., basic level) understanding of the fundamental (i.e., core) elements of literature reviews from a pragmatic perspective. Our article aims to shed light on these issues and hopes to provide clarity for future scholarly endeavors.

Having a foundational understanding of literature reviews as independent studies is (i) *necessary* when addressing the aforementioned issues; (ii) *important* in reconciling and scaffolding our understanding, and (iii) *relevant and timely* due to the proliferation of literature reviews as independent studies. To *contribute* a solution toward addressing this *gap*, we *aim* to demystify review articles as independent studies from a pragmatic standpoint (i.e., practicality). To do so, we deliberately (i) move away from review procedures, protocols, and nomenclatures, and (ii) invest our attention in developing a parsimonious, scaffolded understanding of the fundamental

elements (i.e., types, focuses, considerations, methods, and contributions) of review articles as independent studies.

Three contributions distinguish our article. It is worth noting that pragmatic guides (i.e., foundational knowledge), such as the present one, are not at odds with extant philosophical guides (i.e., advanced knowledge), but rather they complement them. Having a foundational knowledge of the fundamental elements of literature reviews as independent studies is *valuable*, as it can help scholars to (i) gain a good grasp of the fundamental elements of literature reviews as independent studies (*1st contribution*), and (ii) mindfully adopt or adapt existing review procedures, protocols, and nomenclatures to better suit the circumstances of their reviews (e.g., choosing and developing a well-defined review nomenclature, and choosing and reporting on review considerations and steps more parsimoniously) (*2nd contribution*). Therefore, this pragmatic guide serves as (iii) a foundational article (i.e., preparatory understanding) for literature reviews as independent studies (*3rd contribution*). Following this, extant guides using a philosophical approach (i.e., advanced understanding) could be relied upon to make informed review decisions (e.g., adoption, adaptation) in response to the conventions of extant review procedures, protocols, and nomenclatures (Fig. 2).

2 Fundamental elements of literature reviews as independent studies

A foundational understanding of literature reviews as independent studies can be acquired through the appreciation of five fundamental elements – i.e., types, focuses, considerations, methods, and contributions – which are illustrated in Fig. 3 and summarized in the following sections.

2.1 Types

There are two *types* of literature reviews as independent studies: *systematic literature reviews (SLRs)* and *non-systematic literature reviews (non-SLRs)*. It is important to recognize that SLRs and non-SLRs are not review nomenclatures (i.e., names/naming) but rather review types (i.e., classifications).

In particular, SLRs are reviews carried out in a systematic way using an adopted or adapted procedure or protocol to guide data curation and analysis, thus enabling transparent disclosure and replicability (Lim et al. 2022a; Kraus et al. 2020). Therefore, any review nomenclature guided by a systematic methodology is essentially an SLR. The origin of this type of literature review can be traced back to the evidence-based medicine movement in the early 1990s, with the objective being to overcome the issue of inconclusive findings in studies for medical treatments (Boell and Ceez-Kecmanovic 2015).

In contrast, non-SLRs are reviews conducted without any systematic procedure or protocol; instead, they weave together relevant literature based on the critical evaluations and (subjective) choices of the author(s) through a process of discovery and critique (e.g., pointing out contradictions and questioning assertions or beliefs); they

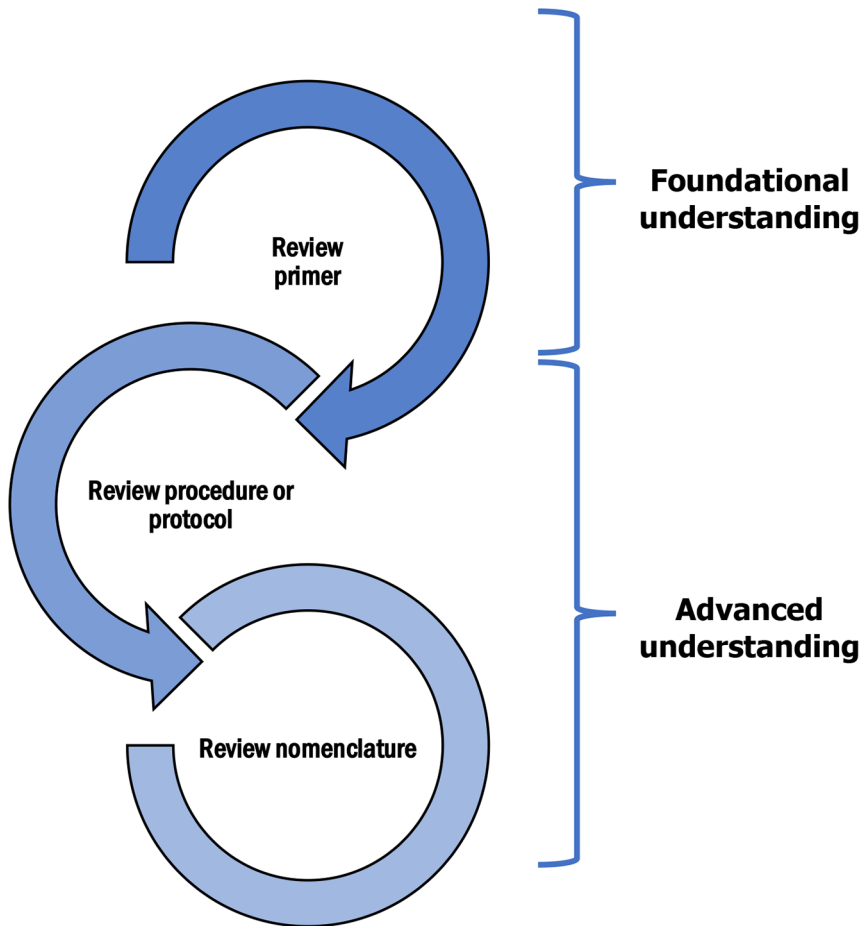


Fig. 2 Foundational and advanced understanding of literature reviews as independent studies

are shaped by the exposure, expertise, and experience (i.e., the “3Es” in judgement calls) of the author(s). Therefore, non-SLRs are essentially critical reviews of the literature (Lim and Weissmann 2021).

2.2 Focuses

Unlike Palmatier et al. (2018) who considered domain-based reviews, theory-based reviews, and method-based reviews as review nomenclatures, we consider *domain*, *theory*, and *method* as three substantive *focuses* that can take center stage in literature reviews as independent studies. This is in line with our attempt to move away from review nomenclatures when providing a foundational understanding of literature reviews as independent studies.

A review that is domain-focused can examine: (i) a *concept* (e.g., customer engagement; Lim et al. 2022b; digital transformation; Kraus et al. 2021; home sharing; Lim

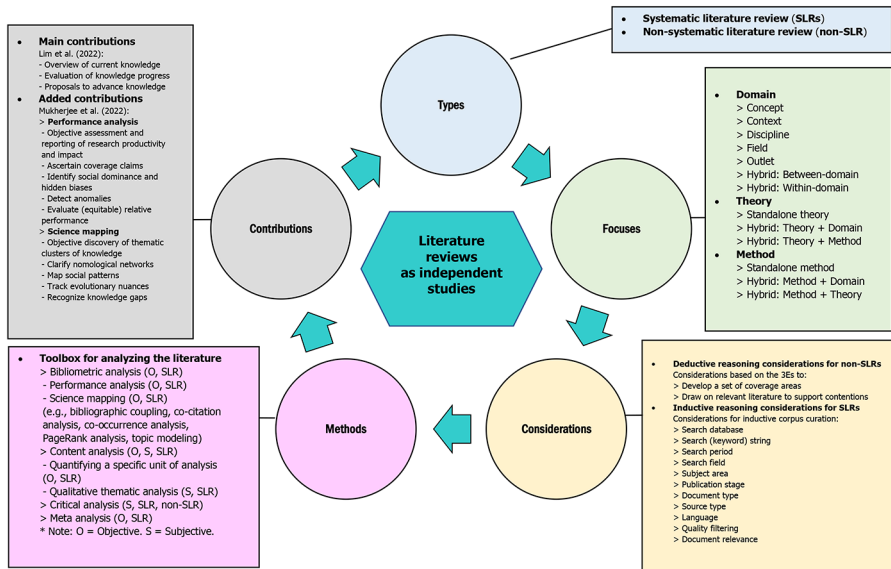


Fig. 3 Fundamental elements of literature reviews as independent studies

et al. 2021; sharing economy; Lim 2020), (ii) a *context* (e.g., India; Mukherjee et al. 2022a), (iii) a *discipline* (e.g., entrepreneurship; Ferreira et al. 2015; international business; Ghauri et al. 2021), (iv) a *field* (e.g., family business; Lahiri et al. 2020; Rovelli et al. 2021; female entrepreneurship; Ojong et al. 2021), or (v) an *outlet* (e.g., *Journal of Business Research*; Donthu et al. 2020; *Management International Review*; Mukherjee et al. 2021; *Review of Managerial Science*; Mas-Tur et al. 2020), which typically offer broad, overarching insights.

Domain-focused *hybrids*, such as the *between-domain* hybrid (e.g., *concept-discipline* hybrid, such as digital transformation in business and management; Kraus et al. 2022; religion in business and entrepreneurship; Kumar et al. 2022a; personality traits in entrepreneurship; Salmony and Kanbach 2022; and policy implications in HR and OB research; Aguinis et al., 2022) and the *within-domain* hybrid (e.g., the *concept-concept* hybrid, such as customer engagement and social media; Lim and Rasul 2022; and global business and organizational excellence; Lim 2022; and the *discipline-discipline* hybrid, such as neuromarketing; Lim 2018) are also common as they can provide finer-grained insights.

A review that is theory-focused can explore a *standalone theory* (e.g., theory of planned behavior; Duan and Jiang 2008), as well as a *theory in conjunction with a domain*, such as the *concept-theory* hybrid (e.g., behavioral control and theory of planned behavior; Lim and Weissmann 2021) and the *theory-discipline* hybrid (e.g., theory of planned behavior in hospitality, leisure, and tourism; Ulker-Demirel and Ciftci 2020), or a *theory in conjunction with a method* (e.g., theory of planned behavior and structural equation modeling).

A review that is method-focused can investigate a *standalone method* (e.g., structural equation modeling; Deng et al. 2018) or a *method in conjunction with a domain*,

such as the *method-discipline* hybrid (e.g., fsQCA in business and management; Kumar et al. 2022b).

2.3 Planning the review, critical considerations, and data collection

The *considerations* required for literature reviews as independent studies depend on their type: SLRs or non-SLRs.

For non-SLRs, scholars often rely on the 3Es (i.e., exposure, expertise, and experience) to provide a critical review of the literature. Scholars who embark on non-SLRs should be well versed with the literature they are dealing with. They should know the state of the literature (e.g., debatable, underexplored, and well-established knowledge areas) and how it needs to be deciphered (e.g., tenets and issues) and approached (e.g., reconciliation proposals and new pathways) to advance theory and practice. In this regard, non-SLRs follow a deductive reasoning approach, whereby scholars initially develop a set of coverage areas for reviewing a domain, theory, or method and subsequently draw on relevant literature to shed light and support scholarly contentions in each area.

For SLRs, scholars often rely on a set of criteria to provide a well-scoped (i.e., breadth and depth), structured (i.e., organized aspects), integrated (i.e., synthesized evidence) and interpreted/narrated (i.e., describing what has happened, how and why) systematic review of the literature.² In this regard, SLRs follow an inductive reasoning approach, whereby a set of criteria is established and implemented to develop a corpus of scholarly documents that scholars can review. They can then deliver a state-of-the-art overview, as well as a future agenda for a domain, theory, or method. Such criteria are often listed in philosophical guides on SLR procedures (e.g., Kraus et al. 2020; Snyder 2019) and protocols (e.g., PRISMA), and they may be adopted/adapted with justifications³. Based on their commonalities they can be summarized as follows:

- *Search database* (e.g., “Scopus” and/or “Web of Science”) can be defined based on justified evidence (e.g., by the two being the largest scientific databases of scholarly articles that can provide on-demand bibliographic data or records; Pranckutė 2021). To avoid biased outcomes due to the scope covered by the selected database, researchers could utilize two or more different databases (Dabić et al. 2021).

² Scoping reviews, structured reviews, integrative reviews, and interpretive/narrative reviews are commonly found in review nomenclature. However, the philosophy of these review nomenclatures essentially reflects what constitutes a good SLR. That is to say, a good SLR should be well scoped, structured, integrated, and interpreted/narrated. This observation reaffirms our position and the value of moving away from review nomenclatures to gain a foundational understanding of literature reviews as independent studies.

³ Given that many of these considerations can be implemented simultaneously in contemporary versions of scientific databases, scholars may choose to consolidate them into a single (or a few) step(s), where appropriate, so that they can be reported more parsimoniously. For a parsimonious but transparent and replicable exemplar, see Lim (2022).

- *Search keywords* may be developed by reading scholarly documents and subsequently brainstorming with experts. The expanding number of databases, journals, periodicals, automated approaches, and semi-automated procedures that use text mining and machine learning can offer researchers the ability to source new, relevant research and forecast the citations of influential studies. This enables them to determine further relevant articles.
- *Boolean operators* (e.g., AND, OR) should be strategically used in developing the *string of search keywords* (e.g., “engagement” AND “customer” OR “consumer” OR “business”). Furthermore, the correct and precise application of quotation marks is important but is very frequently sidestepped, resulting in incorrect selection processes and differentiated results.
- *Search period* (e.g., between a specified period [e.g., 2000 to 2020] or up to the latest full year at the time of writing [e.g., up to 2021]) can be defined based on the justified scope of study (e.g., contemporary evolution versus historical trajectory).
- *Search field* (e.g., “article title, abstract, keywords”) can be defined based on justified assumptions (e.g., it is assumed that the focus of relevant documents will be mentioned in the article title, abstract, and/or keywords).
- *Subject area* (e.g., “business, management, and accounting”) can be defined based on justified principles (e.g., the focus of the review is on the marketing discipline, which is located under the “business, management, and accounting” subject area in Scopus).
- *Publication stage* (e.g., “final”) can be defined based on justified grounds (e.g., enabling greater accuracy in replication).
- *Document type* (e.g., “article” and/or “review”), which reflects the type of scientific/practical contributions (e.g., empirical, synthesis, thought), can be defined based on justified rationales (e.g., articles selected because they are peer-reviewed; editorials not selected because they are not peer-reviewed).
- *Source type* (e.g., “journal”) can be defined based on justified reasons (e.g., journals selected because they publish finalized work; conference proceedings not selected because they are work in progress, and in business/management, they are usually not being considered as full-fledged “publications”).
- *Language* (e.g., “English”) can be determined based on justified limitations (e.g., nowadays, there are not many reasons to use another language besides the academic lingua franca English). Different spellings should also be considered, as the literature may contain both American and British spelling variants (e.g., organization and organisation). Truncation and wildcards in searches are recommended to capture both sets of spellings. It is important to note that each database varies in its symbology.
- *Quality filtering* (e.g., “A*” and “A” or “4*”, “4”, and “3”) can be defined based on justified motivations (e.g., the goal is to unpack the most originally and rigorously produced knowledge, which is the hallmark of premier journals, such as those ranked “A*” and “A” by the *Australian Business Deans Council* [ABDC] Journal Quality List [JQL] and rated “4*”, “4”, and “3” by the Chartered Association of Business Schools [CABS] Academic Journal Guide [AJG]).

- *Document relevance* (i.e., within the focus of the review) can be defined based on justified judgement (e.g., for a review focusing on customer engagement, articles that mention customer engagement as a passing remark without actually investigating it would be excluded).
- *Others: Screening process* should be accomplished by beginning with the deduction of duplicate results from other databases, tracked using abstract screening to exclude unfitting studies, and ending with the full-text screening of the remaining documents.
- *Others: Exclusion-inclusion criteria* interpretation of the abstracts/articles is obligatory when deciding whether or not the articles dealt with the matter. This step could involve removing a huge percentage of initially recognized articles.
- *Others: Codebook building* pertains to the development of a codebook of the main descriptors within a specific field. An inductive approach can be followed and, in this case, descriptors are not established beforehand. Instead, they are established through the analysis of the articles' content. This procedure is made up of several stages: (i) the extraction of important content from titles, abstracts, and keywords; (ii) the classification of this content to form a reduced list of the core descriptors; and (iii) revising the codebook in iterations and combining similar categories, thus developing a short list of descriptors (López-Duarte et al. 2016, p. 512; Dabić et al. 2015; Vlacic et al. 2021).

2.4 Methods

Various methods are used to analyze the pertinent literature. Often, scholars choose a method for corpus analysis before corpus curation. Knowing the analytical technique beforehand is useful, as it allows researchers to acquire and prepare the right data in the right format. This typically occurs when scholars have decided upon and justified pursuing a specific review nomenclature upfront (e.g., bibliometric reviews) based on the problem at hand (e.g., broad domain [outlet] with a large corpus [thousands of articles], such as a premier journal that has been publishing for decades) (Donthu et al. 2021). However, this may not be applicable in instances where (i) scholars do not curate a corpus of articles (non-SLRs), and (ii) scholars only know the size of the corpus of articles once that corpus is curated (SLRs). Therefore, scholars may wish to decide on a method of analyzing the literature depending on (i) whether they rely on a corpus of articles (i.e., yes or no), and (ii) the size of the corpus of articles that they rely on to review the literature (i.e., $n=0$ to ∞).

When analytical techniques (e.g., bibliometric analysis, critical analysis, meta-analysis) are decoupled from review nomenclatures (e.g., bibliometric reviews, critical reviews, meta-analytical reviews), we uncover a toolbox of the following methods for use when analyzing the literature:

- *Bibliometric analysis* measures the literature and processes data by using algorithm, arithmetic, and statistics to analyze, explore, organize, and investigate large amounts of data. This enables scholars to identify and recognize potential “hidden patterns” that could help them during the literature review process.

Bibliometrics allows scholars to objectively analyze a large corpus of articles (e.g., high hundreds or more) using quantitative techniques (Donthu et al. 2021). There are two overarching categories for bibliometric analysis: performance analysis and science mapping. *Performance analysis* enables scholars to assess the productivity (publication) and impact (citation) of the literature relating to a domain, method, or theory using various quantitative metrics (e.g., average citations per publication or year, *h*-index, *g*-index, *i*-index). *Science mapping* grants scholars the ability to map the literature in that domain, method, or theory based on bibliographic data (e.g., *bibliographic coupling* generates thematic clusters based on similarities in shared bibliographic data [e.g., references] among citing articles; *co-citation analysis* generates thematic clusters based on commonly cited articles; *co-occurrence analysis* generates thematic clusters based on bibliographic data [e.g., keywords] that commonly appear together; *PageRank analysis* generates thematic clusters based on articles that are commonly cited in highly cited articles; and *topic modeling* generates thematic clusters based on the natural language processing of bibliographic data [e.g., article title, abstract, and keywords]).⁴ Given the advancement in algorithms and technology, reviews using bibliometric analysis are considered to be *smart* (Kraus et al. 2021) and *technologically-empowered* (Kumar et al. 2022b) SLRs, in which a review has harnessed the benefits of (i) the machine learning of the bibliographic data of scholarly research from technologically-empowered scientific databases, and (ii) big data analytics involving various science mapping techniques (Kumar et al. 2022c).

- *Content analysis* allows scholars to analyze a small to medium corpus of articles (i.e., tens to low hundreds) using quantitative and qualitative techniques. From a *quantitative perspective*, scholars can objectively carry out a content analysis by *quantifying a specific unit of analysis*. A useful method of doing so involves adopting, adapting, or developing an *organizing framework*. For example, Lim et al. (2021) employed an organizing (ADO-TCM) framework to quantify content in academic literature based on: (i) the categories of knowledge; (ii) the relationships between antecedents, decisions, and outcomes; and (iii) the theories, contexts, and methods used to develop the understanding for (i) and (ii). The rapid evolution of software for content analysis allows scholars to carry out complex elaborations on the corpus of analyzed articles, so much so that the most recent software enables the semi-automatic development of an *organizing framework* (Ammirato et al. 2022). From a *qualitative perspective*, scholars can conduct a content analysis or, more specifically, a *thematic analysis*, by subjectively organizing the content into themes. For example, Creevey et al. (2022) reviewed the literature on social media and luxury, providing insights on five core themes (i.e., luxury brand strategy, luxury brand social media communications, luxury consumer attitudes and perceptions, engagement, and the influence of social media

⁴ Where keywords are present (e.g., author keywords or keywords derived from machine learning [e.g., natural language processing]), it is assumed that each keyword represents a specific meaning (e.g., topic [concept, context], method), and that a collection of keywords grouped under the same cluster represents a specific theme.

on brand performance-related outcomes) generated through a content (thematic) analysis. Systematic approaches for inductive concept development through qualitative research are similarly applied in literature reviews in an attempt to reduce the subjectivity of derived themes. Following the principles of the approach by Gioia et al. (2012), Korherr and Kanbach (2021) develop a taxonomy of human-related capabilities in big data analytics. Building on a sample of 75 studies for the literature review, 33 first-order concepts are identified. These are categorized into 15 second-order themes and are finally merged into five aggregate dimensions. Using the same procedure, Leemann and Kanbach (2022) identify 240 idiosyncratic dynamic capabilities in a sample of 34 studies for their literature review. They then categorize these into 19 dynamic sub-capabilities. The advancement of technology also makes it possible to conduct content analysis using computer assisted qualitative data analysis (CAQDA) software (e.g., ATLAS.ti, Nvivo, Quirkos) (Lim et al. 2022a).

- *Critical analysis* allows scholars to subjectively use their 3Es (i.e., exposure, expertise, and experience) to provide a critical evaluation of academic literature. This analysis is typically used in non-SLRs, and can be deployed in tandem with other analyses, such as bibliometric analysis and content analysis in SLRs, which are used to discuss consensual, contradictory, and underexplored areas of the literature. For SLRs, scholars are encouraged to engage in critical evaluations of the literature so that they can truly contribute to advancing theory and practice (Baker et al. 2022; Lim et al. 2022a; Mukherjee et al. 2022b).
- *Meta-analysis* allows scholars to objectively establish a quantitative estimate of commonly studied relationships in the literature (Grewal et al. 2018). This analysis is typically employed in SLRs intending to reconcile a myriad of relationships (Lim et al. 2022a). The relationships established are often made up of conflicting evidence (e.g., a positive or significant effect in one study, but a negative or non-significant effect in another study). However, through meta-analysis, scholars are able to identify potential factors (e.g., contexts or sociodemographic information) that may have led to the conflict.
- *Others: Multiple correspondence analysis* helps to map the field, assessing the associations between qualitative content within a matrix of variables and cases. *Homogeneity Analysis by Means of Alternating Least Squares (HOMALS)* is also considered useful in allowing researchers to map out the intellectual structure of a variety of research fields (Gonzalez-Loureiro et al. 2015; Gonzalez-Loureiro 2021; Obradović et al. 2021). *HOMALS* can be performed in R or used along with a matrix through SPSS software. In summary, the overall objective of this analysis is to discover a low dimensional representation of the original high dimensional space (i.e., the matrix of descriptors and articles). To measure the goodness of fit, a loss function is used. This function is used minimally, and the *HOMALS* algorithm is applied to the least squares loss functions in SPSS. This analysis provides a proximity map, in which articles and descriptors are shown in low-dimensional spaces (typically on two axes). Keywords are paired and each couple that appears together in a large number of articles is shown to be closer on the map and vice-versa.

When conducting a literature review, software solutions allow researchers to cover a broad range of variables, from built-in functions of statistical software packages to software orientated towards meta-analyses, and from commercial to open-source solutions. Personal preference plays a huge role, but the decision as to which software will be the most useful is entirely dependent on how complex the methods and the dataset are. Of all the commercial software providers, we have found the built-in functions of (i) R and VOSviewer most useful in performing bibliometric analysis (Aria and Cuccurullo 2017; R Core Team 2021; Van Eck and Waltman 2014) and (ii) Stata most useful in performing meta-analytical tasks.

Many different analytical tools have been used. These include simple document counting, citation analysis, word frequency analysis, cluster analysis, co-word analysis, and cooperation analysis (Daim et al. 2006). Software has also been produced for bibliometric analysis, such as the Thomson Data Analyzer (TDA), which Thomson Reuters created, and CiteSpace developed by Chen (2013). VOSviewer helps us to construct and visualize bibliometric networks, which can include articles, journals, authors, countries, and institutions, among others (Van Eck and Waltman 2014). These can be organized based on citations, co-citations, bibliographic coupling, or co-authorship relations. In addition, VOSviewer provides text mining functions, which can be used to facilitate a better understanding of co-occurrence networks with regards to the key terms taken from a body of scientific literature (Donthu et al. 2021; Wong 2018). Other frequently used tools include for bibliometric analysis include Bibliometrix/Biblioshiny in R, CitNetExplorer, and Gephi, among others.

2.5 Contributions

Well-conducted literature reviews may make multiple contributions to the literature as standalone, independent studies.

Generally, there are three primary contributions of literature reviews as independent studies: (i) to provide *an overview of current knowledge* in the domain, method, or theory, (ii) to provide *an evaluation of knowledge progression* in the domain, method, or theory, including the establishment of key knowledge, conflicting or inconclusive findings, and emerging and underexplored areas, and (iii) to provide *a proposal for potential pathways for advancing knowledge* in the domain, method, or theory (Lim et al. 2022a, p. 487). Developing theory through literature reviews can take many forms, including organizing and categorizing the literature, problematizing the literature, identifying and exposing contradictions, developing analogies and metaphors, and setting out new narratives and conceptualizations (Breslin and Gatrell 2020). Taken collectively, these contributions offer crystalized, evidence-based insights that both ‘mine’ and ‘prospect’ the literature, highlighting extant gaps and how they can be resolved (e.g., flags paradoxes or theoretical tensions, explaining why something has not been done, what the challenges are, and how these challenges can be overcome). These contributions can be derived through successful bibliometric analysis, content analysis, critical analysis, and meta-analysis.

Additionally, the deployment of specific methods can bring in further added value. For example, a performance analysis in a bibliometric analysis can contribute to: (i) *objectively assessing and reporting research productivity and impact*; (ii) *ascertain*

taining reach for coverage claims; (iii) identifying social dominance and hidden biases; (iv) detecting anomalies; and (v) evaluating (equitable) relative performance; whereas science mapping in bibliometric analysis can contribute to: (i) objectively discovering thematic clusters of knowledge; (ii) clarifying nomological networks; (iii) mapping social patterns; (iv) tracking evolutionary nuances; and (v) recognizing knowledge gaps (Mukherjee et al. 2022b, p. 105).

3 Conclusion

Independent literature reviews will continue to be written as a result of their necessity, importance, relevance, and urgency when it comes to advancing knowledge (Lim et al. 2022a; Mukherjee et al. 2022b), and this can be seen in the increasing number of reviews being published over the last several years. Literature reviews advance academic discussion. Journal publications on various topics and subject areas are becoming more frequent sites for publication. This trend will only heighten the need for literature reviews. This article offers directions and control points that address the needs of three different stakeholder groups: producers (i.e., potential authors), evaluators (i.e., journal editors and reviewers), and users (i.e., new researchers looking to learn more about a particular methodological issue, and those teaching the next generation of scholars). Future producers will derive value from this article's teachings on the different fundamental elements and methodological nuances of literature reviews. Procedural knowledge (i.e., using control points to assist in decision-making during the manuscript preparation phase) will also be of use. Evaluators will be able to make use of the procedural and declarative knowledge evident in control points as well. As previously outlined, the need to cultivate novelty within research on business and management practices is vital. Scholars must also be supported to choose not only safe mining approaches; they should also be encouraged to attempt more challenging and risky ventures. It is important to note that abstracts often seem to offer a lot of potential, stating that authors intend to make large conceptual contributions, broadening the horizons of the field.

Our article offers important insights also for practitioners. Noteworthy, our framework can support corporate managers in decomposing and better understanding literature reviews as ad-hoc and independent studies about specific topics that matter for their organization. For instance, practitioners can understand more easily what are the emerging trends within their domain of interest and make corporate decisions in line with such trends.

This article arises from an intentional decoupling from philosophy, in favor of adopting a more pragmatic approach. This approach can assist us in clarifying the fundamental elements of literature reviews as independent studies. Five fundamental elements must be considered: types, focuses, considerations, methods, and contributions. These elements offer a useful frame for scholars starting to work on a literature review. Overview articles (guides) such as ours are thus invaluable, as they equip scholars with a solid foundational understanding of the integral elements of a literature review. Scholars can then put these teachings into practice, armed with a

better understanding of the philosophy that underpins the procedures, protocols, and nomenclatures of literature reviews as independent studies.

Funding Open access funding provided by Libera Università di Bolzano within the CRUI-CARE Agreement.

Data availability Our manuscript has no associate data.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Aguinis H, Jensen SH, Kraus S (2022) Policy implications of organizational behavior and human resource management research. *Acad Manage Perspect* 36(3):1–22
- Ammirato S, Felicetti AM, Rogano D, Linzalone R, Corvello V (2022) Digitalising the systematic literature review process: The My SLR platform. *Knowl Manage Res Pract*. doi:<https://doi.org/10.1080/14778238.2022.2041375>
- Aria M, Cuccurullo C (2017) bibliometrix: An R-tool for comprehensive science mapping analysis. *J Informetrics* 11(4):959–975
- Baker WE, Mukherjee D, Perin MG (2022) Learning orientation and competitive advantage: A critical synthesis and future directions. *J Bus Res* 144:863–873
- Boell SK, Cecez-Kecmanovic D (2015) On being 'systematic' in literature reviews. *J Inform Technol* 30:161–173
- Borrego M, Foster MJ, Froyd JE (2014) Systematic literature reviews in engineering education and other developing interdisciplinary fields. *J Eng Educ* 103(1):45–76
- Breslin D, Gatrell C (2020) Theorizing through literature reviews: The miner-prospecter continuum. *Organizational Res Methods*. <https://doi.org/10.1177/1094428120943288> (in press)
- Cajal B, Jiménez R, Gervilla E, Montaña JJ (2020) Doing a systematic review in health sciences. *Clínica y Salud* 31(2):77–83
- Chen C (2013) Mapping scientific frontiers: The quest for knowledge visualization. Springer Science & Business Media
- Creevey D, Coughlan J, O'Connor C (2022) Social media and luxury: A systematic literature review. *Int J Manage Reviews* 24(1):99–129
- Dabić M, González-Loureiro M, Harvey M (2015) Evolving research on expatriates: what is 'known' after four decades (1970–2012). *Int J Hum Resource Manage* 26(3):316–337
- Dabić M, Vlačić B, Kiessling T, Caputo A, Pellegrini M (2021) Serial entrepreneurs: A review of literature and guidance for future research. *Journal of Small Business Management*, 1–36
- Daim TU, Rueda G, Martin H, Gerdts P (2006) Forecasting emerging technologies: Use of bibliometrics and patent analysis. *Technol Forecast Soc Chang* 73(8):981–1012
- Deng L, Yang M, Marcoulides KM (2018) Structural equation modeling with many variables: A systematic review of issues and developments. *Front Psychol* 9:580
- Donthu N, Kumar S, Pattnaik D (2020) Forty-five years of Journal of Business Research: A bibliometric analysis. *J Bus Res* 109:1–14
- Donthu N, Kumar S, Mukherjee D, Pandey N, Lim WM (2021) How to conduct a bibliometric analysis: An overview and guidelines. *J Bus Res* 133:285–296
- Duan W, Jiang G (2008) A review of the theory of planned behavior. *Adv Psychol Sci* 16(2):315–320

- Durach CF, Kembro J, Wieland A (2017) A new paradigm for systematic literature reviews in supply chain management. *J Supply Chain Manage* 53(4):67–85
- Fan D, Breslin D, Callahan JL, Szatt-White M (2022) Advancing literature review methodology through rigour, generativity, scope and transparency. *Int J Manage Reviews* 24(2):171–180
- Ferreira MP, Reis NR, Miranda R (2015) Thirty years of entrepreneurship research published in top journals: Analysis of citations, co-citations and themes. *J Global Entrepreneurship Res* 5(1):1–22
- Ghauri P, Strange R, Cooke FL (2021) Research on international business: The new realities. *Int Bus Rev* 30(2):101794
- Gioia DA, Corley KG, Hamilton AL (2012) Seeking qualitative rigor in inductive research: Notes on the gioia methodology. *Organizational Res Methods* 16(1):15–31
- Gonzalez-Loureiro M, Dabić M, Kiessling T (2015) Supply chain management as the key to a firm's strategy in the global marketplace: Trends and research agenda. *Int J Phys Distribution Logistics Manage* 45(1/2):159–181. <https://doi.org/10.1108/IJPDLM-05-2013-0124>
- Grewal D, Puccinelli N, Monroe KB (2018) Meta-analysis: Integrating accumulated knowledge. *J Acad Mark Sci* 46(1):9–30
- Hansen C, Steinmetz H, Block J (2021) How to conduct a meta-analysis in eight steps: a practical guide. *Management Review Quarterly*, 1–19
- Korherr P, Kanbach DK (2021) Human-related capabilities in big data analytics: A taxonomy of human factors with impact on firm performance. *RMS*. <https://doi.org/10.1007/s11846-021-00506-4> (in press)
- Kraus S, Breier M, Dasi-Rodríguez S (2020) The art of crafting a systematic literature review in entrepreneurship research. *Int Entrepreneurship Manage J* 16(3):1023–1042
- Kraus S, Durst S, Ferreira J, Veiga P, Kailer N, Weinmann A (2022) Digital transformation in business and management research: An overview of the current status quo. *Int J Inf Manag* 63:102466
- Kraus S, Jones P, Kailer N, Weinmann A, Chaparro-Banegas N, Roig-Tierno N (2021) Digital transformation: An overview of the current state of the art of research. *Sage Open* 11(3):1–15
- Kraus S, Mahto RV, Walsh ST (2021) The importance of literature reviews in small business and entrepreneurship research. *J Small Bus Manage*. <https://doi.org/10.1080/00472778.2021.1955128> (in press)
- Kumar S, Sahoo S, Lim WM, Dana LP (2022a) Religion as a social shaping force in entrepreneurship and business: Insights from a technology-empowered systematic literature review. *Technol Forecast Soc Chang* 175:121393
- Kumar S, Sahoo S, Lim WM, Kraus S, Bamel U (2022b) Fuzzy-set qualitative comparative analysis (fsQCA) in business and management research: A contemporary overview. *Technol Forecast Soc Chang* 178:121599
- Kumar S, Sharma D, Rao S, Lim WM, Mangla SK (2022c) Past, present, and future of sustainable finance: Insights from big data analytics through machine learning of scholarly research. *Ann Oper Res*. <https://doi.org/10.1007/s10479-021-04410-8> (in press)
- Laher S, Hassem T (2020) Doing systematic reviews in psychology. *South Afr J Psychol* 50(4):450–468
- Leemann N, Kanbach DK (2022) Toward a taxonomy of dynamic capabilities – a systematic literature review. *Manage Res Rev* 45(4):486–501
- Lahiri S, Mukherjee D, Peng MW (2020) Behind the internationalization of family SMEs: A strategy tripod synthesis. *Glob Strategy J* 10(4):813–838
- Lim WM (2018) Demystifying neuromarketing. *J Bus Res* 91:205–220
- Lim WM (2020) The sharing economy: A marketing perspective. *Australasian Mark J* 28(3):4–13
- Lim WM (2022) Ushering a new era of Global Business and Organizational Excellence: Taking a leaf out of recent trends in the new normal. *Global Bus Organizational Excellence* 41(5):5–13
- Lim WM, Rasul T (2022) Customer engagement and social media: Revisiting the past to inform the future. *J Bus Res* 148:325–342
- Lim WM, Weissmann MA (2021) Toward a theory of behavioral control. *J Strategic Mark*. <https://doi.org/10.1080/0965254X.2021.1890190> (in press)
- Lim WM, Kumar S, Ali F (2022a) Advancing knowledge through literature reviews: 'What', 'why', and 'how to contribute'. *Serv Ind J* 42(7–8):481–513
- Lim WM, Rasul T, Kumar S, Ala M (2022b) Past, present, and future of customer engagement. *J Bus Res* 140:439–458
- Lim WM, Yap SF, Makkar M (2021) Home sharing in marketing and tourism at a tipping point: What do we know, how do we know, and where should we be heading? *J Bus Res* 122:534–566

- López-Duarte C, González-Loureiro M, Vidal-Suárez MM, González-Díaz B (2016) International strategic alliances and national culture: Mapping the field and developing a research agenda. *J World Bus* 51(4):511–524
- Mas-Tur A, Kraus S, Brandtner M, Ewert R, Kürsten W (2020) Advances in management research: A bibliometric overview of the Review of Managerial Science. *RMS* 14(5):933–958
- Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA (2015) Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Reviews* 4(1):1–9
- Mukherjee D, Kumar S, Donthu N, Pandey N (2021) Research published in Management International Review from 2006 to 2020: A bibliometric analysis and future directions. *Manage Int Rev* 61:599–642
- Mukherjee D, Kumar S, Mukherjee D, Goyal K (2022a) Mapping five decades of international business and management research on India: A bibliometric analysis and future directions. *J Bus Res* 145:864–891
- Mukherjee D, Lim WM, Kumar S, Donthu N (2022b) Guidelines for advancing theory and practice through bibliometric research. *J Bus Res* 148:101–115
- Obradović T, Vlačić B, Dabić M (2021) Open innovation in the manufacturing industry: A review and research agenda. *Technovation* 102:102221
- Ojong N, Simba A, Dana LP (2021) Female entrepreneurship in Africa: A review, trends, and future research directions. *J Bus Res* 132:233–248
- Palmatier RW, Houston MB, Hulland J (2018) Review articles: Purpose, process, and structure. *J Acad Mark Sci* 46(1):1–5
- Post C, Sarala R, Gatrell C, Prescott JE (2020) Advancing theory with review articles. *J Manage Stud* 57(2):351–376
- Pranckutė R (2021) Web of Science (WoS) and Scopus: The titans of bibliographic information in today's academic world. *Publications* 9(1):12
- R Core Team (2021) R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL <https://www.R-project.org/> Accessed 20th July 2022
- Rovelli P, Ferasso M, De Massis A, Kraus S (2021) Thirty years of research in family business journals: Status quo and future directions. *Journal of Family Business Strategy*, 100422
- Salmony FU, Kanbach DK (2022) Personality trait differences across types of entrepreneurs: a systematic literature review. *RMS* 16:713–749
- Snyder H (2019) Literature review as a research methodology: An overview and guidelines. *J Bus Res* 104:333–339
- Tranfield D, Denyer D, Smart P (2003) Towards a methodology for developing evidence-informed management knowledge by means of systematic review. *Br J Manag* 14(3):207–222
- Ulker-Demirel E, Ciftci G (2020) A systematic literature review of the theory of planned behavior in tourism, leisure and hospitality management research. *J Hospitality Tourism Manage* 43:209–219
- Van Eck NJ, Waltma L (2014) CitNetExplorer: A new software tool for analyzing and visualizing citation networks. *J Informetrics* 8(4):802–823
- Vlačić B, Corbo L, Silva e, Dabić M (2021) The evolving role of artificial intelligence in marketing: A review and research agenda. *J Bus Res* 128:187–203
- Wong D (2018) VOSviewer. *Tech Serv Q* 35(2):219–220. <https://doi.org/10.1080/07317131.2018.1425352>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Sascha Kraus^{1,2} · Matthias Breier³ · Weng Marc Lim^{4,8,22} · Marina Dabić^{5,6} · Satish Kumar^{7,8} · Dominik Kanbach^{9,10} · Debmalya Mukherjee¹¹ · Vincenzo Corvello¹² · Juan Piñeiro-Chousa¹³ · Eric Liguori¹⁴ · Daniel Palacios-Marqués¹⁵ · Francesco Schiavone^{16,17} · Alberto Ferraris^{18,21} · Cristina Fernandes^{19,20} · João J. Ferreira¹⁹

- ✉ Sascha Kraus
sascha.kraus@zfke.de
- Matthias Breier
matthias.breier@gmx.at
- Weng Marc Lim
lim@wengmarc.com
- Marina Dabić
mdabic@net.efzg.hr
- Satish Kumar
skumar.dms@mnit.ac.in
- Dominik Kanbach
D.Kanbach@hhl.de
- Debmalya Mukherjee
dmukher@uakron.edu
- Vincenzo Corvello
vincenzo.corvello@unime.it
- Juan Piñeiro-Chousa
j.pineiro@usc.es
- Eric Liguori
liguori@rowan.edu
- Daniel Palacios-Marqués
dapamar@doe.upv.es
- Francesco Schiavone
francesco.schiavone@uniparthenope.it
- Alberto Ferraris
alberto.ferraris@unito.it
- Cristina Fernandes
cristina.isabel.fernandes@ubi.pt
- João J. Ferreira
jjmf@ubi.pt

- ¹ Faculty of Economics & Management, Free University of Bozen-Bolzano, Bolzano, Italy
- ² Department of Business Management, University of Johannesburg, Johannesburg, South Africa
- ³ School of Business, Lappeenranta University of Technology, Lappeenranta, Finland
- ⁴ Sunway University Business School, Sunway University, Sunway City, Malaysia
- ⁵ Faculty of Economics and Business, University of Zagreb, Zagreb, Croatia
- ⁶ School of Economics and Business, University of Ljubljana, Ljubljana, Slovenia
- ⁷ Department of Management Studies, Malaviya National Institute of Technology Jaipur, Jaipur, India
- ⁸ Faculty of Business, Design and Arts, Swinburne University of Technology, Kuching, Malaysia
- ⁹ Chair of Strategic Management and Digital Entrepreneurship, HHL Leipzig Graduate School of Management, Leipzig, Germany
- ¹⁰ School of Business, Woxsen University, Hyderabad, India

- ¹¹ College of Business, The University of Akron, Akron, USA
- ¹² Department of Engineering, University of Messina, Messina, Italy
- ¹³ Department of Finance, Santiago de Compostela University, Santiago de Compostela, Spain
- ¹⁴ Rowan University, Rohrer College of Business, Glassboro, NJ, USA
- ¹⁵ School of Engineering Design, Universidad Politécnica de Valencia, Valencia, Spain
- ¹⁶ Department of Management and Quantitative Studies, Parthenope University, Naples, Italy
- ¹⁷ Paris School of Business, Paris, France
- ¹⁸ Department of Management, University of Turin, Turin, Italy
- ¹⁹ Department of Management and Economics & NECE Research Unit in Business Sciences, University of Beira Interior, Covilha, Portugal
- ²⁰ Centre for Corporate Entrepreneurship and Innovation, Loughborough University, Loughborough, UK
- ²¹ Laboratory for International and Regional Economics, Graduate School of Economics and Management, Ural Federal University, Yekaterinburg, Russia
- ²² School of Business, Law and Entrepreneurship, Swinburne University of Technology, Hawthorn, Australia